SRP 16-BAND UPLAM WITH MASK R-CNN FOR AUDIOVISUAL SELD

Technical Report

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ABSTRACT

We present an audiovisual sound event localization and detection system based on a modified Mask R-CNN with a ResNet-50 Feature Pyramid Network backbone, taking 19-channel inputs composed of 3 RGB video channels and 16 UpLAM acoustic feature maps extracted from the tetrahedral microphone array. The model is trained with progressive backbone unfreezing, class-balanced sampling with inverse-frequency weights (cap=30), focal loss, and data augmentation including azimuth rotation, horizontal flip, RGB dropout, and audio band masking.

***Index Terms*—** One, two, three, four, five

1. INTRODUCTION

DCASE 2026 Task 3 advances sound event localization and detection (SELD) through semantic acoustic imaging, focusing on the integration of spatial audio and audiovisual scenes. This challenge emphasizes the development of robust systems capable of identifying and localizing sound events in complex acoustic environments.

This report outlines the technical approach for participating in challange with the primary aim of redesigning the UpLam model to operate with 16 perceptually relevant Mel frequency bands and leveraging AudibleLight for scalable synthetic data generation.

2. SYSTEM ARCHITECTURE AND MODIFICATIONS

Iwan: baseline default UpLam and modyfied Mask R-CNN

Rafal: modyfied UpLam

3. SYNTHETIC DATA GENERATION

Given the growing use of synthetic data across different machine learning tasks and the limited amount of data in the original STARSS23 dataset, we chose to use AudibleLight to generate additional data.

AudibleLight is a Python package that provides a unified API for generating synthetic soundscapes through ray-traced simulations and parametric room impulse responses (RIRs). It integrates frameworks like SpatialScaper, SoundSpaces, and Pyroomacoustics to ensure scalable acoustic diversity.

For the generation process, we set parameters to include 5 to 20 static events per 60-second scene, with no moving events. The maximum overlap was set to 4, and the reference decibel level was -50 dB. Events were positioned between 0.5 to 3 meters from the microphone, with a height limit of 0.3 meters relative to the microphone position, which was randomly chosen within a randomly selected mesh. We used the Eigenmike32 mic array, with video resolution set to 1920 x 960 pixels at 10 fps. There was a 50% chance of adding noise, randomly selected from available options, and the video low power mode was disabled to maintain higher quality meshes.

After generation, the audio was resampled to 24 kHz and 4 appropriate tetrahedral channels were selected. The video was resized to 360:180 resolution, with frames saved as PNG files to meet task specifications.This process produced 122 recordings, totaling in 2 hours and 2 minutes of synthetic data, which were then combined with the original train split of the STARSS23 dataset.

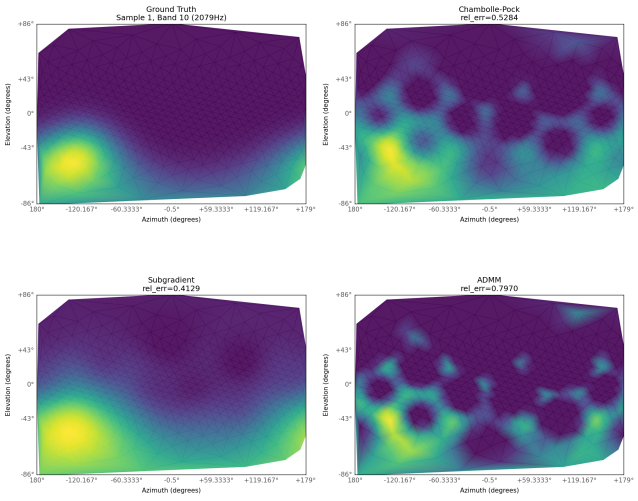


Figure 1: Acoustic map reconstructions on LOCATA data. Chambolle-Pock and ADMM converge to the exact minimizer but exhibit checkerboard artifacts. Subgradient descent and the APGD ground truth remain smooth due to early stopping.

4. A NOTE ON NUMERICAL APPROACHES TO LAM

The LAM pipeline relies on APGD-computed acoustic intensity maps as ground truth for training. We investigated whether exact numerical solvers (Chambolle-Pock, subgradient descent, ADMM) could produce improved estimates. Figure 1 shows representative reconstructions on real LOCATA Eigenmike recordings.

ADMM successfully recovers the dominant acoustic source, confirming the mathematical validity of the imaging operator. How-

ever, the Hessian $H_{ij} = |a_i^H a_j|^2$ is singular: with 32 microphones mapping onto 484 spatial directions at sub-wavelength resolution, the system is fundamentally underdetermined. Adding quadratic regularization renders the problem well-posed, enabling convergence within tens of iterations.

The true minimizer of the loss, however, exhibits characteristic checkerboard and ringing artifacts (Fig. 1). ADMM, as the solver that achieves the lowest objective value, reveals these artifacts most clearly. The APGD-based ground truth used in the LAM pipeline produces smooth, visually plausible maps precisely because it is stopped early, before reaching this minimizer. The regularizer promotes piecewise constancy and sparsity but does not encode the physical structure of compact, contiguous acoustic sources on the sphere.

This has implications for training: the LAM model learns to replicate APGD's early-stopped estimates, which are smooth approximations to a loss function whose exact solution does not correspond to the desired acoustic map.

A further consequence is that early stopping precludes warm-starting across consecutive frames. In principle, since acoustic scenes change slowly, the solution from one time frame provides an excellent initialization for the next. This would amortize solver cost: rather than running hundreds of iterations per frame, a handful of fine-tuning steps would suffice, making exact solvers practical for streaming applications. However, if we rely on early stopping to suppress artifacts, warm-starting becomes self-defeating: it accelerates convergence toward the exact minimizer, which is precisely what early stopping was designed to avoid. Maintaining both a running estimate and artifact-free maps would require a frame-dependent stopping rule calibrated to the distance already traveled—an open problem under the current objective.

5. EXPERIMENTAL METHODOLOGY AND RESULTS

Rafal: training process, parameters, unfreezing etc. GPU? epoch time?
Rafal F: modyfications in inference
Evaluation dataset
Metrics
metrics

6. CONCLUSION AND FUTURE DIRECTIONS

Bogdan: UpLam training approach
more bands in uplamm good!
synthetic data maybe good
low mAP, ok pearson?

or list them by yourself

7. REFERENCES

[1] http://www.cs.tut.fi/sgn/arg/dcase2016/.

[2] PDF specification for IEEE Xplore, http://www.ieee.org/portal/cms_docs/pubs/confstandards/pdfs/IEEE-PDF-SpecV401.pdf.

[3] Creating high resolution PDF files for book production with open source tools, http://www.grassbook.org/neteler/highres_pdf.html.

[4] E. Williams, *Fourier Acoustics: Sound Radiation and Nearfield Acoustic Holography*. London, UK: Academic Press, 1999.

[5] http://www.ieee.org/web/publications/rights/copyrightmain.html.

[6] C. Jones, A. Smith, and E. Roberts, "A sample paper in conference proceedings," in *Proc. IEEE ICASSP*, vol. II, 2003, pp. 803–806.

[7] A. Smith, C. Jones, and E. Roberts, "A sample paper in journals," *IEEE Trans. Signal Process.*, vol. 62, pp. 291–294, Jan. 2000.

Table 1: Example Table with Macro and Micro Evaluation Metrics

Run Name	Metric	AP@0.25	AP@0.5	AP@0.75	mAP	Pearson r
Baseline	Macro	0.0010	0.0000	0.0000	0.0003	0.1358
	Micro	0.0032	0.0000	0.0000	0.0011	0.4429
SRP-UPL-MRCNN	Macro	**0.0277**	**0.0069**	**0.0001**	**0.0116**	0.2796
	Micro	0.0279	0.0072	**0.0001**	0.0117	0.4270
SRP-UPL-MRCNN-2	Macro	0.0025	0.0000	0.0000	0.0009	**0.3455**
	Micro	0.0112	0.0000	0.0000	0.0037	0.4189
SRP-UPL-MRCNN-3	Macro	0.0212	0.0046	0.0000	0.0086	0.3287
	Micro	**0.0308**	**0.0080**	0.0000	**0.0129**	**0.4791**
SRP-UPL-MRCNN-3	Macro	0.0012	0.0000	0.0000	0.0004	0.3352
	Micro	0.0025	0.0001	0.0000	0.0009	0.4624